

AWS Automated Cloud Resource & Security Reporter

1. Project Overview

This project is a serverless AWS automation solution that automatically collects AWS resource and security information every day and sends a detailed report through email.

The project uses:

- 1) AWS Lambda
- 2) Amazon EventBridge Scheduler
- 3) Amazon SES (Simple Email Service)
- 4) IAM
- 5) CloudWatch
- 6) Python (Boto3)

The main purpose of this project is to reduce manual AWS monitoring and provide a daily summary of cloud resources and security status.

2. Problem Statement

In AWS environments, administrators need to regularly check:

- How many EC2 instances are running
- Which S3 buckets exist
- Which IAM users exist
- Whether MFA is enabled
- Security status of resources

Doing this manually every day is time-consuming.

To solve this problem, I built an automated reporting system.

3. Architecture

EventBridge Scheduler

|

| Daily 8:00 AM

v

AWS Lambda Function

|

+----> EC2 Information

|

+----> S3 Information

|

+----> IAM Information

|

+----> MFA Status Check

|

v

Generate Report

|

v

Amazon SES

|

v

Email Delivery

4. AWS Services Used

a) AWS Lambda

Lambda is the core service of the project.

Purpose:

- Runs Python code
- Collects AWS resource information
- Creates the report
- Sends email

Why Lambda?

- No server management
- Pay only when executed
- Fully automated
- Easy integration with AWS services

b) Amazon EventBridge Scheduler

Purpose:

Automatically triggers Lambda

Current Configuration:

Schedule Name:

DailyCloudResourceReport

Time:

8:00 AM IST

Frequency:

Daily

Every morning at 8 AM:

EventBridge



Starts Lambda

No manual execution is required.

c) Amazon SES

Purpose:

* Sends email reports

SES receives the report generated by Lambda and sends it to:

xyz@gmail.com

Email Subject: AWS Daily Resource & Security Report

d) IAM

Purpose:

* Secure access control

Lambda role permissions:

- AmazonEC2ReadOnlyAccess
- AmazonS3ReadOnlyAccess
- IAMReadOnlyAccess
- AmazonSESEFullAccess
- AWSLambdaBasicExecutionRole

These permissions allow Lambda to:

- Read EC2 details
- Read S3 details
- Read IAM details
- Send emails
- Write logs

e) CloudWatch

Purpose:

- Monitoring
- Logging
- Troubleshooting

Whenever Lambda runs:

- Execution logs
- Errors
- Execution duration
- Memory usage

are automatically stored in CloudWatch.

5. Working Flow

Step 1

Every day at 8:00 AM:

EventBridge Scheduler

automatically starts the Lambda function.

Step 2

Lambda begins execution.

Python code runs using Boto3.

Boto3 is AWS SDK for Python.

It allows Python code to communicate with AWS services.

Step 3

Lambda connects to EC2.

python

```
ec2.describe_instances()
```

It collects:

- Instance ID
- Instance State
- Instance Type
- Public IP Address

Example:

Instance ID:

i-0abc123456

State:

running

Instance Type:

t2.micro

Public IP:

13.233.xx.xx

Step 4

Lambda connects to S3.

```
python
```

```
s3.list_buckets()
```

It collects:

- Bucket names
- Total bucket count

Example:

Bucket Name:

project-backup

Bucket Name:

website-data

Step 5

Lambda connects to IAM.

```
python
```

```
iam.list_users()
```

It collects:

- IAM username
- User details

Example:

awswithlix

admin-user

Step 6

Lambda checks MFA status.

python

iam.list_mfa_devices()

For every IAM user:

text

MFA Enabled

or

MFA Disabled

Example:

text

User:

awswithlix

MFA:

Enabled

This provides a security audit.

Step 7

Lambda generates a report.

Example:

AWS DAILY RESOURCE & SECURITY REPORT

EC2 INSTANCES

Instance ID:

i-12345

State:

running

Instance Type:

t2.micro

Public IP:

13.233.xx.xx

Total EC2:

3

S3 BUCKETS

project-backup

Total Buckets:

1

IAM USERS

awswithlix

MFA:

Enabled

Total Users:

1

Step 8

Lambda sends the report to Amazon SES.

python

ses.send_email()

SES processes the request.

Step 9

Amazon SES sends the email.

Destination:

xyz@gmail.com

Step 10

The report arrives in *Gmail* inbox.

Daily report example:

Subject:

AWS Daily Resource & Security Report

Users can review cloud resources without logging into *AWS Console*.

6. Security Features

This project also performs a basic security audit.

Checks:

- IAM MFA Status
- Enabled
- Disabled
- IAM User Information
- Username
- Last Password Used
- Resource Visibility
- EC2 Instances
- S3 Buckets
- IAM User
- This helps identify security gaps.

7. Advantages of the Project

- Automation
- No manual checking required.
- Serverless
- No EC2 server needed.
- Cost Effective
- Lambda runs only once daily.
- Security Monitoring
- MFA status tracking.
- Centralized Reporting
- All information in one email.
- Scalable
- Additional AWS services can be added easily.

8. Future Enhancements

In future, this project can be extended to include:

CloudWatch Metrics

- CPU Utilization
- Memory Usage
- Disk Usage